

A quality-aware pipeline for ranking GWAS summary statistics for reproducible PRS construction

Candidate GWAS files



- GWAS Catalog summary statistics
- 284 candidate files
- 13 phenotypes
- Multiple files per phenotype
- Compressed and uncompressed formats

Schema harmonisation



- Raw headers detected
- Headers normalised
- Standard GWAS columns mapped
- Alleles and effect sizes standardised
- FinalGWAS.csv generated

Feature extraction



- F1 File structure
- F2 Column mapping
- F3 Harmonisation
- F4 Variant quality
- F5 Statistical features
- F6 LDSC heritability
- F7 Reference-panel overlap
- F8 Target compatibility
- F9 PRS-readiness

Target overlap assessment



- Target PLINK BIM compared
- rsID overlap calculated
- Coordinate overlap calculated
- Allele-compatible SNPs retained
- Usable variants counted

PRS construction



- PLINK C+T scoring
- PRSice-2 scoring
- Train folds evaluated
- Test folds evaluated
- PRS performance recorded

Machine learning ranking



- Nested feature sets trained
- Model 1 uses F1-F6
- Model 2 uses F1-F8
- Model 3 uses F1-F9
- LOPO validation applied
- Final GWAS ranking produced